

Stephen Kioni Mwangi

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EXPERIENCE

Canonical <i>Software Engineer</i>	January 2024 - Present <i>Remote</i>
- Designed and implemented a bespoke message queue service for remote device management, enabling reliable async communication between the cloud and remote IoT/edge devices at scale (<i>Golang, Kubernetes</i>) - Implemented client-side device management features in snapd (Ubuntu's package manager daemon), enabling remote configuration and control of IoT devices running snapd (<i>Golang</i>) - Authored technical specifications for the remote device management system, covering the messaging protocol, authorization model, message queue architecture, and APIs - Built out the Landscape client snap from the ground up, shipping user management, configuration management, service management, and zero-touch provisioning which enabled management of Ubuntu Core devices (<i>Python</i>)	
Savannah Informatics <i>Software Engineer</i>	June 2022 - December 2023 <i>Nairobi, Kenya</i>
- Led the design and development of the backend of an Electronic Medical Records and Patient Engagement system (<i>Python, Django, Golang, PostgreSQL, Kubernetes, Google Cloud Platform, RabbitMQ</i>) - Implemented and designed the company's SMS and USSD API gateway and integrated it with a major Kenyan telecommunications provider, reducing communication costs by $\approx 40\%$ - Migrated my team's backend infrastructure from Google Cloud Run to Google Kubernetes Engine, reducing infrastructure costs by $\approx 55\%$ - Implemented an event-driven usage-based billing module that integrated with prepaid wallets on an internal ERP - Optimized existing APIs through application-level caching and query optimization which reduced p95 latency by $\approx 80\%$ (Talk given on Django performance)	
Savannah Informatics <i>Software Engineer Intern</i>	June 2021 - May 2022 <i>Nairobi, Kenya</i>
Maintained and developed the Slade360 Electronic Data Interchange system (<i>Python, Django</i>)	

EDUCATION

University of Nairobi <i>BSc. Computer Science</i>	September 2018 - September 2022 <i>First Class Honours</i>
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PROJECTS

Obsidian Spaced Repetition	April 2021 - Present
Flashcards and spaced repetition plugin written in TypeScript for the note-taking tool Obsidian . 500,000+ downloads, translated into 15+ languages by a global community of users.	
clstr.io	December 2025 - Present
An open-source platform for learning distributed systems by building them from scratch, with guided challenges where users validate their implementations against a test harness that orchestrates clusters and simulates network conditions.	
Sourcerer MCP	August 2025 - Present
An MCP server for semantic code search using Tree-sitter and vector embeddings, letting AI agents jump directly to relevant code chunks instead of reading entire files, reducing token waste.	

SKILLS

Languages:	Golang, Python (Django), SQL, TypeScript, JavaScript
Infrastructure:	Kubernetes, Google Cloud Platform, Docker, PostgreSQL, Redis, RabbitMQ
Observability:	Grafana, Sentry, Loki, Tempo
CI/CD:	GitHub Actions, GitLab CI, Helm, Ansible